ABSTRACT OF THE DISCLOSURE

A system for the automated physical transport of data, comprising an autonomous robotic unit equipped with a manipulator and an interface configured to handle a portable data cartridge having a high-bandwidth optical interface. The system solves the logistical challenges of physical data transport in high-performance environments. The robotic unit, which may be an aerial drone, a ground vehicle, or a stationary robotic arm, autonomously retrieves a data cartridge from a data port, transports it to a designated destination, and docks it with a destination port. This architecture enables fully automated, dynamically reconfigurable physical data fabrics within data centers and facilitates rapid, secure physical data logistics between remote facilities without requiring human intervention.